

ISTANBUL ATATURK, Turkey

WMO: 170600

Lat: 40.967N

Lon: 28.817E

Elev: 108

StdP: 14.64

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.5	31.9	20.1	15.2	36.0	23.0	17.4	37.1	30.1	35.0	27.6	37.1	14.6	0	0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.0	89.7	70.7	87.7	70.3	85.7	70.0	76.1	82.4	74.6	81.7	73.3	80.9	12.6	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	126.5	79.7	72.0	118.7	78.9	70.7	113.3	78.3	39.6	82.1	38.2	81.3	37.0	81.2	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.0	22.3	20.1	DB	24.8	94.8	4.2	2.9	21.7	96.9	19.3	98.6	16.9	100.2	13.8	102.3	(4)
(5)			WB	23.3	77.3	4.0	2.2	20.4	78.9	18.0	80.2	15.8	81.4	12.8	83.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.4	43.7	44.7	48.2	55.6	64.7	73.3	78.3	78.7	71.4	62.7	54.6	47.2
	DBStd	13.70	6.65	6.54	6.31	6.17	5.31	4.75	3.55	3.80	5.06	5.49	6.46	6.53
	HDD50	674	212	170	112	17	0	0	0	0	0	1	30	132
	HDD65	3089	660	569	520	288	70	2	0	0	8	107	314	551
	CDD50	4454	17	20	56	185	457	698	878	889	643	396	169	46
	CDD65	1396	0	0	0	6	62	250	413	424	201	37	3	0
	CDH74	9970	0	0	0	23	235	1544	3513	3633	956	64	2	0
	CDH80	2712	0	0	0	2	31	375	1061	1083	156	4	0	0
Wind	WSAvg	10.0	10.7	10.7	9.9	8.9	8.8	9.3	10.7	11.0	9.7	9.7	9.4	11.0
Precipitation	PrecAvg	25.2	3.7	2.4	2.3	2.0	1.2	1.1	0.9	0.9	1.2	2.2	3.3	4.4
	PrecMax	40.4	7.6	5.4	5.2	6.5	4.5	2.3	3.3	3.7	6.4	5.5	8.2	14.2
	PrecMin	17.8	0.5	0.5	0.6	0.4	0.1	0.0	0.0	0.0	0.0	0.1	1.4	0.8
	PrecStd	5.0	2.0	1.3	1.4	1.3	0.9	0.7	0.8	0.9	1.3	1.5	1.6	2.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.5	62.4	68.1	77.2	83.8	90.0	93.2	93.0	87.7	79.2	72.0	65.8
		MCWB	51.8	51.9	54.0	59.7	65.3	69.4	69.3	70.7	68.5	66.0	59.3	55.8
	2%	DB	57.4	58.7	64.1	71.8	79.2	86.7	89.7	89.7	84.1	75.4	68.4	61.0
		MCWB	50.3	50.5	52.9	57.9	63.3	68.4	70.4	71.5	67.7	64.9	59.1	54.3
	5%	DB	55.3	55.8	60.7	68.3	76.6	84.2	87.6	87.7	81.0	73.0	66.0	58.8
		MCWB	49.4	49.4	51.8	56.4	62.8	67.9	70.1	71.3	67.6	63.7	58.4	53.0
	10%	DB	53.2	53.7	57.4	65.4	73.5	81.8	85.6	85.7	79.0	70.2	63.3	56.5
		MCWB	48.3	48.4	50.2	55.0	61.8	67.1	70.0	70.9	67.0	62.7	57.1	51.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.5	53.6	56.0	61.3	68.4	74.0	78.0	78.3	73.7	69.3	62.9	57.3
		MCDB	58.2	59.7	64.8	72.5	77.6	81.8	82.5	83.6	81.1	74.9	67.8	63.0
	2%	WB	51.4	51.7	53.9	59.2	66.0	72.3	76.0	76.5	71.7	67.1	60.7	55.1
		MCDB	55.7	57.0	61.5	68.5	74.5	80.1	81.8	83.0	78.4	72.3	66.3	60.0
	5%	WB	50.0	50.2	52.3	57.5	64.6	71.0	74.6	75.0	70.2	65.3	59.1	53.4
		MCDB	54.0	55.0	59.1	66.3	73.0	79.0	81.3	82.4	76.9	70.8	64.3	58.0
	10%	WB	48.5	48.7	50.9	55.9	63.4	69.7	73.2	73.7	69.2	63.7	57.5	51.8
		MCDB	52.8	53.3	56.8	63.6	71.4	78.2	80.9	81.2	76.1	69.3	62.7	56.0
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.6	11.2	13.4	13.7	13.9	13.7	13.0	12.5	10.4	9.8	8.4
		MCDBR	10.8	12.4	15.8	18.3	17.7	17.0	16.1	15.2	14.5	13.2	12.3	9.5
		MCWBR	6.7	7.3	7.9	8.5	7.8	7.2	7.1	6.0	6.2	6.3	6.4	6.2
	5% WB	MCDBR	9.8	11.7	14.0	16.4	15.0	14.6	13.8	13.1	13.2	11.9	10.8	9.2
		MCWBR	7.0	7.5	7.9	8.5	7.7	7.1	7.1	6.0	6.5	6.5	6.6	6.7
Clear-Sky Solar Irradiance	taub	0.372	0.403	0.439	0.480	0.443	0.433	0.436	0.445	0.434	0.415	0.385	0.376	
	taud	2.313	2.206	2.123	2.028	2.209	2.282	2.285	2.260	2.274	2.294	2.324	2.306	
	Ebn at Noon	241	250	255	253	266	268	266	260	254	244	234	228	
	Edh at Noon	29	37	45	53	45	42	41	41	38	34	28	27	
All-Sky Solar Radiation	RadAvg	469	684	1098	1539	1961	2213	2275	1963	1449	943	611	430	
	RadStd	69	100	104	167	147	163	75	101	106	86	82	63	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+1.35	N/A	N/A	+1.06	N/A	N/A	-95	-254	+308	+153
(47) Regional (13 neighbors)	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	-176	N/A	+79

Nomenclature: See separate page